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MAT 213 Brief Calculus

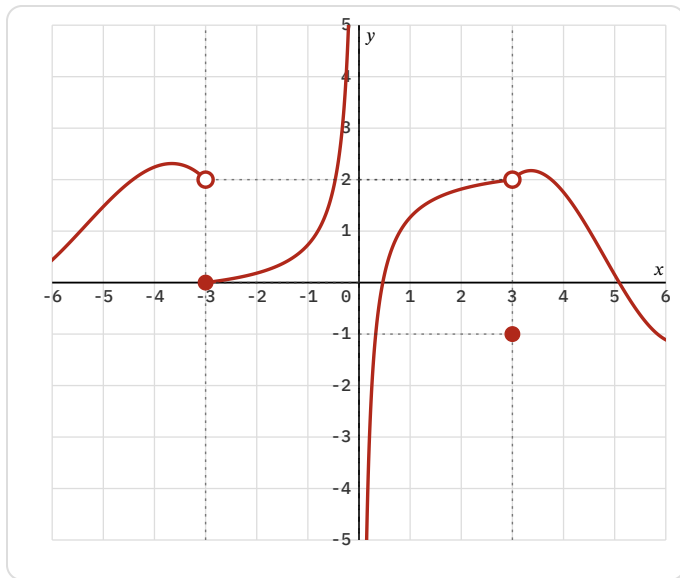
Mock Midterm C — Answer Key

Practice midterm (AI-Free) · Sections 1.1–1.4

Total: 60 points

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE.



a.) $\lim_{x \rightarrow 0^-} f(x)$ (2)

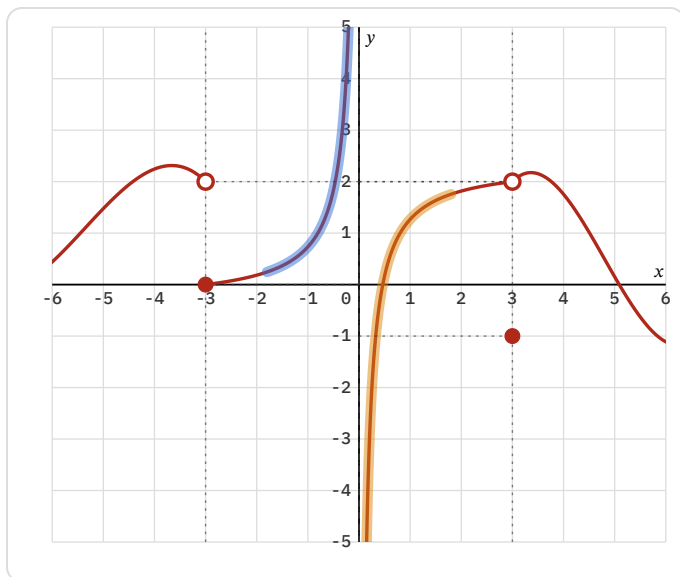
∞

b.) $f(3)$ (2)

-1

4 pts

WORKED SOLUTION



— approaching $x = 0$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

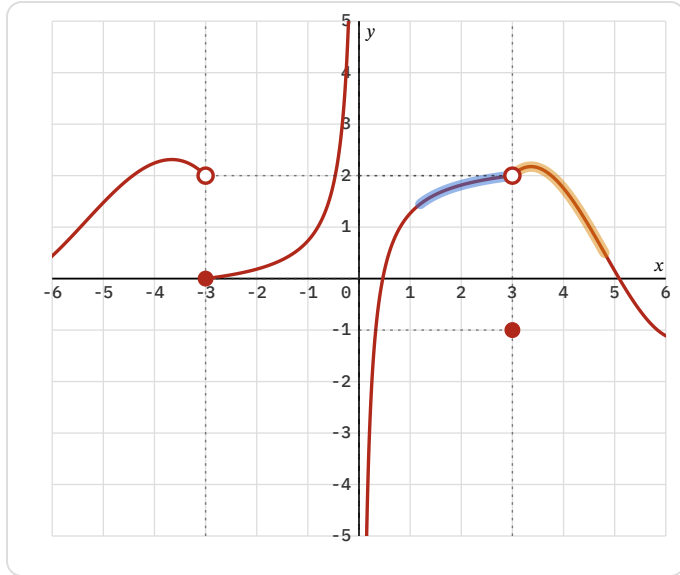
At $x = 0$

There is no filled dot on the line $x = 0$, so $f(0)$ is undefined $\rightarrow$ DNE.

From the left, the curve runs along the asymptote $x = 0$ upward, so the limit is ∞ .

From the right, the curve runs along the asymptote $x = 0$ downward, so the limit is $-\infty$.

The left and right limits don't match, so $\lim_{x \rightarrow 0} f(x)$ DNE.



— approaching $x = 3$ from the left — from the right. Follow each colored piece of the curve to the height it heads toward.

At $x = 3$

The **filled** dot at $x = 3$ sits at height -1, so $f(3) = -1$.

Tracing the curve toward $x = 3$ from the left, the heights approach 2 (open or closed dot doesn't matter).

Tracing the curve toward $x = 3$ from the right, the heights approach 2 (open or closed dot doesn't matter).

Both one-sided limits equal 2, so $\lim_{x \rightarrow 3} f(x) = 2$ — even though $f(3) = -1$. The limit ignores the value AT the point.

2. Let $f(x) = \frac{-2}{(x-3)^2}$. Answer with a number, ∞ , $-\infty$, or DNE. 4 pts
- a.) $\lim_{x \rightarrow 3^-} f(x)$ (2)
 $-\infty$
- b.) The vertical asymptote is $x =$ (2)
 3

WORKED SOLUTION

Near $x = 3$ the denominator $(x - 3)^2$ is tiny, so $f(x)$ is huge in size. Only the sign needs work.

Squaring makes the denominator a small **positive** number on both sides, so both one-sided limits are $-\infty$. Since both sides agree we write $\lim_{x \rightarrow 3} f(x) = -\infty$ (it still has no finite value).

As $x \rightarrow \infty$, the denominator grows without bound while the top stays -2, so $f(x) \rightarrow 0$. The vertical asymptote is $x = 3$.

3. Suppose 4 pts
- $\lim_{x \rightarrow 3} f(x) = -3, \quad \lim_{x \rightarrow 3} g(x) = 4, \quad \lim_{x \rightarrow 3} h(x) = 9$
- Use the properties of limits to find each limit.
- a.) $\lim_{x \rightarrow 3} x^2 f(x)$ (2)
 -27
- b.) $\lim_{x \rightarrow 3} [3f(x) + g(x)]^2$ (2)
 25

WORKED SOLUTION

a. $\lim_{x \rightarrow 3} x^2 f(x) = (3)^2 \cdot (-3) = -27$

b. $\lim_{x \rightarrow 3} \frac{f(x)g(x)}{h(x)+4} = \frac{(-3)(4)}{9+4} = -\frac{12}{13}$

c. $\lim_{x \rightarrow 3} \sqrt{h(x)} f(x) = \sqrt{9} \cdot (-3) = 3 \cdot (-3) = -9$

d. $\lim_{x \rightarrow 3} [3f(x) + g(x)]^2 = [3(-3) + (4)]^2 = (-5)^2 = 25$

The quotient property works here because the bottom limit isn't 0.

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow 36} \frac{\sqrt{x}-6}{x-36}$ (2)

$\frac{1}{12}$

b.) $\lim_{x \rightarrow -2} \frac{3x^2+4x-4}{x^2+6x+8}$ (2)

-4

4 pts

WORKED SOLUTION

a.

$$\lim_{x \rightarrow 36} \frac{\sqrt{x}-6}{x-36} = \lim_{x \rightarrow 36} \frac{\sqrt{x}-6}{(\sqrt{x}-6)(\sqrt{x}+6)} = \lim_{x \rightarrow 36} \frac{1}{\sqrt{x}+6} = \frac{1}{6+6} = \frac{1}{12}$$

b.

$$\lim_{x \rightarrow -2} \frac{3x^2+4x-4}{x^2+6x+8} = \lim_{x \rightarrow -2} \frac{(x+2)(3x-2)}{(x+2)(x+4)} = \lim_{x \rightarrow -2} \frac{3x-2}{x+4} = \frac{-8}{2} = -4$$

Notation tip: keep writing **lim** on every line until the step where you actually substitute.

5. Compute each limit algebraically. Use $\infty / -\infty$ for unbounded results.

4 pts

a.) $\lim_{x \rightarrow \infty} \frac{-3x+7}{-2x^2-9x}$ (2)

0

b.) $\lim_{x \rightarrow \infty} \frac{4x^3+6x^2+6x}{-5x^3-8x^2+1}$ (2)

$-\frac{4}{5}$

WORKED SOLUTION

a. Divide every term by x^2 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{-3x+7}{-2x^2-9x} = \lim_{x \rightarrow \infty} \frac{-\frac{3}{x} + \frac{7}{x^2}}{-2 - \frac{9}{x}}$$

The top $\rightarrow 0$ while the bottom $\rightarrow -2$, so the limit is 0.

b. Divide every term by x^2 , the highest power of x in the denominator:

$$\lim_{x \rightarrow -\infty} \frac{-x^3}{6x^2+6x} = \lim_{x \rightarrow -\infty} \frac{-x}{6 + \frac{6}{x}}$$

The bottom $\rightarrow 6$, but the top is dominated by $-1x^1$, which $\rightarrow \infty$ as $x \rightarrow -\infty$; dividing by 6 gives ∞ .

c. Divide every term by x^3 , the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{4x^3+6x^2+6x}{-5x^3-8x^2+1} = \lim_{x \rightarrow \infty} \frac{4 + \frac{6}{x} + \frac{6}{x^2}}{-5 - \frac{8}{x} + \frac{1}{x^3}}$$

Every fraction term $\rightarrow 0$, leaving $\frac{4}{-5} = -\frac{4}{5}$.

6. Find each limit (end behavior). Use ∞ or $-\infty$ when the outputs grow without bound. 4 pts

a.) $\lim_{x \rightarrow -\infty} (-4x^5 + 5x^2 + 3x)$ (2)

∞

b.) $\lim_{x \rightarrow -\infty} (-e^{-x})$ (2)

$-\infty$

WORKED SOLUTION

a. Leading term $-4x^5$: degree 5 (odd), leading coefficient negative. As $x \rightarrow -\infty$, x^5 is negative, times -4 $\rightarrow \infty$.

b. Leading term $-2x^5$: degree 5 (odd), leading coefficient negative. As $x \rightarrow -\infty$, x^5 is negative, times -2 $\rightarrow \infty$.

c. $\frac{8}{x^3} \rightarrow 0$ as $x \rightarrow -\infty$, leaving -3.

d. As $x \rightarrow -\infty$, e^{-x} grows without bound, and multiplying by -1 gives $-\infty$.

7. Let

4 pts

$$f(x) = \begin{cases} \frac{2x^2 + 15x + 25}{2x + 5} & x < -\frac{5}{2} \\ -2x - 3 & x \geq -\frac{5}{2} \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 1} f(x)$ (2)
 -5

b.) $\lim_{x \rightarrow -\frac{5}{2}} f(x)$ (2)
DNE

WORKED SOLUTION

For $x < -\frac{5}{2}$, factor the top: $\frac{(2x+5)(x+5)}{2x+5} = x + 5$ (valid since $x \neq -\frac{5}{2}$ there).

a. $1 > -\frac{5}{2} \rightarrow$ bottom piece: $-2(1) - 3 = -5$.

b. $-5 < -\frac{5}{2} \rightarrow$ top piece, simplified: $-5 + 5 = 0$.

c. The bottom piece includes $x = -\frac{5}{2}$ ($\geq$), so $f(-\frac{5}{2}) = -2(-\frac{5}{2}) - 3 = 2$.

d. Left: $\lim_{x \rightarrow -\frac{5}{2}-} (x + 5) = \frac{5}{2}$.

e. Right: $\lim_{x \rightarrow -\frac{5}{2}+} (-2x - 3) = 2$.

f. $\frac{5}{2} \neq 2$, so the limit DNE.

8. Find the value of k that makes f continuous at $x = 1$.

4 pts

$$f(x) = \begin{cases} x^2 - x & x < 1 \\ kx - 6 & x \geq 1 \end{cases}$$

a.) $\lim_{x \rightarrow 1^-} f(x)$ (2)
 0

b.) $k =$ (2)
 6

WORKED SOLUTION

Left: $(1)^2 - 1(1) = 0$. Right (and $f(1)$): $k(1) - 6$.

$$1k - 6 = 0 \Rightarrow k = \frac{6}{1} = 6$$

9.

Classify each function at the given x -value.

2 pts

$$\text{a.) } f(x) = \begin{cases} x - 2 & x < 2 \\ x - 4 & x \geq 2 \end{cases} \text{ at } x = 2 \quad (1)$$

(C) Jump discontinuity

$$\text{b.) } f(x) = \frac{x+2}{x^2+6x+8} \text{ at } x = -4 \quad (1)$$

(D) Infinite discontinuity

WORKED SOLUTIONa. Left $\rightarrow 0$, right $\rightarrow -2$. Different finite values $\rightarrow$ jump.b. $0/0$ at 4 , but it cancels to $x + 4$, so the limit exists (8) while $f(4)$ is undefined $\rightarrow$ removable.c. Factor: $\frac{(x+2)}{(x+2)(x+4)}$. The $(x + 4)$ factor does NOT cancel, so $x = -4$ is a vertical asymptote $\rightarrow$ infinite.

10.

Let $f(x) = x^2 + 6x + 5$, where $f(x)$ represents the number of miles driven after x hours. Find the average rate of change over $[2, 6]$.

$$\text{a.) } f(2) \quad (2)$$

21

$$\text{b.) } \text{Which sentence correctly states the result with units?} \quad (1)$$

(D) Between $x = 2$ and $x = 6$ hours, the distance driven changes at an average rate of 14 miles per hour.

3 pts

WORKED SOLUTION

$$f(2) = 21, \quad f(6) = 77$$

$$\frac{f(6) - f(2)}{6 - 2} = \frac{77 - (21)}{4} = \frac{56}{4} = 14 \text{ miles per hour}$$

It's an *average* over an interval (not an instant), and the units are miles per hour — miles per hour. The total change would be 56 miles.

11. The table shows the U.S. unemployment rate.

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Rate, %	4.6	5.1	5.6	5.4	6.1	6.5	5.9	5.6	5.7	6.0	6.0

Rounded answers (2+ decimal places) are accepted.

- a.) Average rate of change from 2017 to 2020 (2)

$$\frac{1}{3}$$

- b.) The units of these answers are... (1)

(A) percentage points per year

3 pts

WORKED SOLUTION

2017–2020:

$$\frac{6.1 - 5.1}{2020 - 2017} = \frac{1}{3} \approx 0.333$$

2023–2026:

$$\frac{6.0 - 5.6}{2026 - 2023} = \frac{0.4}{3} \approx 0.133$$

Units: output units per input unit → percentage points per year.

12.

Let $f(x) = -x^2 + 4x - 5$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) (3)

$$-2x - h + 4$$

b.) As $h \rightarrow 0$, what value do these approach? (2)

$$0$$

5 pts

WORKED SOLUTION

$$f(x+h) = -(x+h)^2 + 4(x+h) - 5 = -x^2 - 2xh - h^2 + 4x + 4h - 5$$

$$f(x+h) - f(x) = -2xh - h^2 + 4h$$

$$\frac{f(x+h) - f(x)}{h} = \frac{h(-2x - h + 4)}{h} = -2x - h + 4$$

Now substitute:

$$x = 2, h = 1 : \quad -1$$

$$x = 2, h = 0.1 : \quad -\frac{1}{10} = -0.1$$

$$x = 2, h = 0.01 : \quad -\frac{1}{100} = -0.01$$

Letting $h \rightarrow 0$ leaves 0: the instantaneous rate of change (slope of the tangent line) at $x = 2$. That's exactly what section 1.4 formalizes.

13.

Let $f(x) = \frac{1}{x}$. Use the **limit definition** $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ (3)

$$\frac{-1}{x^2}$$

b.) What does $f'(1) = -1$ mean? (1)

(C) The instantaneous rate of change of f at $x = 1$ is -1 — the slope of the tangent line there.

4 pts

WORKED SOLUTION

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \lim_{h \rightarrow 0} \frac{\frac{1x-1(x+h)}{x(x+h)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1h}{x(x+h)} \cdot \frac{1}{h} = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} = \frac{-1}{x^2} \end{aligned}$$

Then $f'(1) = -1$ and the slope of the tangent at $x = 2$ is $f'(2) = -\frac{1}{4}$.

$f'(1)$ is the *instantaneous* rate of change at one input — the slope of the tangent line, not the height of the graph and not an average over an interval.

14. (a) Suppose $f(1) = -3$ and $f'(1) = -1$. Write the equation of the tangent line to f at $x = 1$.
- (b) The tangent line to the graph of g at $x = -1$ is $y = -3x - 3$. Find $g(-1)$ and $g'(-1)$.
- a.) (a) Tangent line (3)
- $$y = -x - 2$$
- b.) (b) $g'(-1)$ (2)
- $$-3$$

5 pts

WORKED SOLUTION

(a) Point $(1, -3)$, slope -1:

$$y + 3 = -(x - 1) \Rightarrow y = -x - 2$$

(b) The tangent line touches g at $x = -1$, so $g(-1)$ is the line's height there:
 $-3(-1) - 3 = 0$. Its slope is the derivative: $g'(-1) = -3$.

15.

Let $h(t) = -16t^2 + 160t$ be the height, in feet, of a rocket t seconds after launch.

a.) $h'(t) =$ (3)
 $-32t + 160$

b.) Interpret $h'(2) = 96$. (1)

(A) At $t = 2$ seconds, the height of the rocket is increasing at a rate of 96 feet per second.

4 pts

WORKED SOLUTION

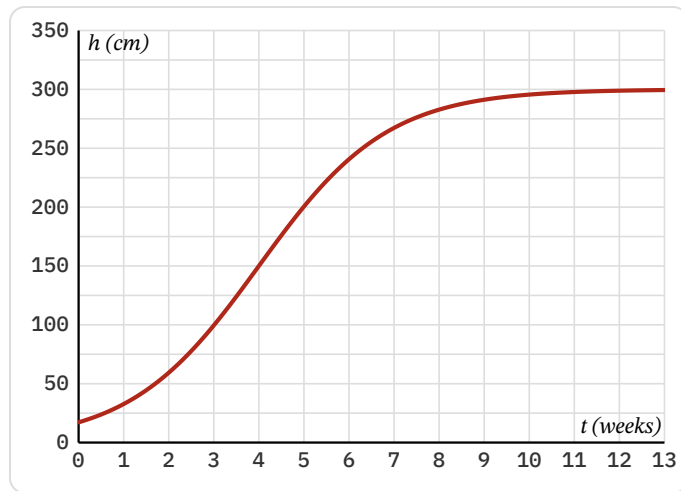
$$\begin{aligned} h'(t) &= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \rightarrow 0} \frac{[-16(t+h)^2 + 160(t+h)] - (-16t^2 + 160t)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-16t^2 - 32th - 16h^2 + 160t + 160h + 16t^2 - 160t}{h} \\ &= \lim_{h \rightarrow 0} \frac{-32th - 16h^2 + 160h}{h} = \lim_{h \rightarrow 0} \frac{h(-32t - 16h + 160)}{h} \\ &= \lim_{h \rightarrow 0} (-32t - 16h + 160) = -32t + 160 \end{aligned}$$

$$h'(2) = -32(2) + 160 = 96$$

Positive/negative tells the direction, and the units are feet per second: at that instant the height of the rocket is increasing at 96 feet per second. (The value $h(2) = 256$ is a different quantity.)

16.

A sunflower seedling is planted and its height h (in centimeters) after t weeks is graphed below. Decide whether each statement is true or false.



- a.) The slopes of the tangent lines get steeper up to $t = 7$, and then the slopes start to flatten out. (1)

False

- b.) The average rate of change over $[10, 12]$ is less than over $[1, 3]$. (1)

True

2 pts

WORKED SOLUTION

a. **False.** The graph is steepest at about $t = 4$ (where it changes from bending up to bending down).

b. **False.** $\text{ARC} \approx \frac{295.6 - 17.2}{10} \approx 27.8$ cm/week; the tangent slope at $t = 3$ is about 46.6 cm/week.

c. **True.** Tangent slopes: about 33.3 cm/week at $t = 2$ vs. 11.3 at $t = 8$.

d. **True.** $[10, 12]: \approx 1.7$ cm/week; $[1, 3]: \approx 33.4$ cm/week.
